

FOCUSED REVISION

Genetic counseling and screening of consanguineous couples and their offspring practice resource: Focused Revision

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Abstract

There are no evidence-based guidelines to inform genetic counseling for consanguineous couples and their offspring. This focused revision builds on the expert opinions from the original publication of "Genetic Counseling and Screening of Consanguineous Couples and Their Offspring," based on a review of literature published since 2002.

KEYWORDS

consanguinity, genetic counselling, genetic screening, genetic testing, incest, newborn screening, SNP microarray, tandem mass spectrometry

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2 | PURPOSE

In 2015, the NSGC Practice Guidelines Committee (PGC) adopted Institute of Medicine (IOM) standards for development of evidence-based clinical practice guidelines. NSGC guidelines published prior to 2015 do not meet IOM criteria to remain classified as a Clinical Practice Guideline. Based on recommendations from expert reviewers, the PGC has concluded that the original Consanguinity Practice Guidelines document, published in 2002 (Bennett et al., 2002), is still relevant and current, except for specific outdated content. Therefore, the PGC has reclassified the Consanguinity Practice Guidelines as a Practice Resource with the addition of the Focused Revision detailed below. A Focused Revision is a brief addendum that focuses on content identified as outdated in the previously published document, without changing the main purpose or intent of the Practice Resource (Kalia & James, 2020).

3 | INTRODUCTION

Screening and testing options available for prenatal, pediatric, and carrier screening have changed since 2002 when the original article was published. The following information provides a focused update on these technologies and practices as they apply to consanguineous couples and their offspring.

4 | POINTS TO CONSIDER

4.1 | Risks associated with consanguinity

Published studies continue to show evidence of an increased risk for birth defects, major medical conditions, and reproductive mortality in the offspring of first cousin unions, mostly attributed to autosomal recessive disorders. (Bennett et al., 2002; Bittles & Black, 2010; Hamamy et al., 2011; Zlotogora & Shalev, 2010). In couples with no known genetic disorders in the family, there is an additional 1.7%-2.8% risk for significant birth defects in the offspring of first cousins above the general population risk. Multinational studies found that among the progeny of first cousin pregnancies, there was a mean excess of 1.1% neonatal and infant deaths compared with the non-consanguineous population (Bittles & Black, 2010). We identified no new studies on offspring of incestuous relationships (refer to table 4, original document).

4.2 | Genetic screening and testing for consanguineous couples and their offspring

Obtaining a 3-4 generation medical family history continues to be an important assessment tool for consanguineous couples. With the increasing recognition of monogenic risk factors for common diseases, such as cancer and cardiovascular disease, genetic counseling for

consanguineous couples and their offspring should include assessment of adult onset conditions with consideration of the potential reproductive risks from homozygosity of pathogenic variants associated with these conditions. Homozygous pathogenic variants in genes typically considered to be autosomal dominant can present with more severe phenotypes or clinically distinct autosomal recessive disorders. Examples include pediatric cardiomyopathy with *ALPK3* (Almomani et al., 2016), constitutional mismatch repair deficiency with Lynch syndrome genes (Bakry et al., 2014), and Fanconi anemia with *BRCA1*, *BRCA2*, *BRIP1*, *PALB2*, and *RAD51C* (Niraj et al., 2019).

No additional evidence-based recommendations for screening or testing consanguineous couples and their offspring were identified; however, advances in technology offer additional clinical testing options that may improve detection of autosomal recessive disorders. Testing approaches may include:

4.2.1 | Expanded carrier screening

Efficiencies in technologies have reduced genetic testing costs and increased the scope of genes that can be concurrently analyzed (Beauchamp et al., 2019; Westemeyer et al., 2020). In a study of a highly consanguineous population Monies et al., (2019) reported that unique variants accounted for more than half of the pathogenic variants identified. Use of screening panels that use sequencing techniques rather than targeted variant testing will improve detection of couples at 25% risk to have offspring with an autosomal recessive condition. (Mor-Shakee et al., 2021; Sallevelt et al., 2021).

4.2.2 | SNP microarray

Chromosomal microarray is now routinely used in the evaluation of abnormal ultrasound findings and for the genetic evaluation of children with intellectual disability, autism, and/or multiple congenital anomalies. In addition to copy-number variants, SNP microarray platforms can identify long stretches of homozygosity indicative of consanguinity. When an abnormality or a suspected genetic disorder is present in the offspring/fetus of a couple with known consanguinity, SNP array can help to target candidate regions of homozygosity and make possible a strategized approach to focused diagnostic testing for recessive genes (Kearney, 2011; Wierenga, 2013).

4.2.3 | Clinical exome/genome sequencing

The use of exome or genome sequencing for offspring of consanguineous couples presenting with a suspected genetic disorder may offer improved diagnostic yield by identifying pathogenic and likely pathogenic variants in rare autosomal recessive disorders (Al-Dewik et al., 2019; Monies et al., 2019; Sallevelt et al., 2021). Due

to shared ancestry, homozygous variants of uncertain significance should be interpreted with caution. Identification of co-inheritance in this context often lacks utility in variant interpretation.

4.2.4 | Newborn screening

Advances in newborn screening practices, including the use of tandem mass spectrometry and routine hearing screening, have increased the diagnostic yield of autosomal recessive disorders. When newborn screening in a healthy infant is performed following the Recommended Uniform Screening Panel (RUSP), no evidence was identified to support offering supplemental newborn screening even when the parents are consanguineous. Consider screening along RUSP recommendations for newborns whose parents are consanguineous if the child did not have standard testing via tandem mass spectrometry and/or hearing screening.

4.3 | Ethical issues and special considerations

Detection of consanguineous relationships has increased with the ability to identify regions of homozygosity using microarray and genomic sequencing. These findings are often addressed in genetic counseling sessions and can present a significant challenge for clinicians, especially when a close biological relationship was not previously known or reported. Pre-test counseling should include the possibility of detecting unanticipated biological relationships.

Speculation of specific biological relationships in laboratory reports is discouraged (Deignan et al., 2020; Rehder et al., 2013). If a test result raises the possibility of incest, the relationship should be confirmed and not assumed. Despite advances in laboratory technologies, there remains no standard approach to genetic counseling in these situations (Sund & Rehder, 2014). Counseling should be tailored to the individual based on family and social history, as well as testing indication. Social work services and an ethics or legal consultation may provide additional guidance (Grote et al., 2014).

There are now fewer states in the United States with laws restricting first cousin marriages. The unique laws in each state continue to necessitate the need for genetic counselors and other health professionals to maintain general knowledge about the laws in their own state and neighboring states in their service area.

In Table 5, the term "intellectual disability" should be used for patient interview questions in place of "mental retardation".

5 | SUMMARY

While there are still no evidence-based guidelines to inform genetic counseling for consanguineous couples and their offspring, this focused revision builds on the expert opinions from the original publication, based on a review of literature published since 2002.

CONFLICT OF INTEREST

Bennett RL, Malleda NR, Byers PH, and Barr KB, declare that they have no conflict of interest. Steiner, R.D. has no specific conflict of interest with the content of this manuscript. Potential conflicts of interest are: equity interest in and consulting fees from Acer Therapeutics and PTC Therapeutics. In the past 5 years, served as a consultant with compensation for Retrophin. Principal investigator of an investigator-initiated observational research study funded by Alexion via contract with Marshfield Clinic Health System. In the past five years, received travel support from Pfizer for a meeting to review clinical trial results, and consulting fees from Raptor, Biomarin, Alexion, E-Scape Bio, Health Advances, Precision for Value, and Best Doctors. Honoraria from Medscape and The France Foundation. Employee of PreventionGenetics.

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